ENVIRONMENTAL ISSUE TRACKING SYSTEM

A software-based solution for campus sustainability

PROJECT TYPE	COMPLEXITY	HARDWARE
Web application	Moderate	Not required

1. Introduction

Environmental issues such as overflowing dustbins, water leakage, littering, and unnecessary electricity consumption can affect campus cleanliness and sustainability. These problems may remain unresolved when there is no centralized platform to report issues, assign responsibility, and monitor resolution.

2. Problem Statement

Environmental issues on campus may remain unresolved because there is no centralized system for reporting problems, assigning responsibility, tracking progress, and monitoring resolution. This can lead to delayed action, recurring problems, and inefficient management of campus environmental resources.

3. Proposed Solution

Develop a web-based platform through which students and staff can report environmental problems and monitor their status. Administrators can review reports, assign responsibility, prioritize issues, and update progress until resolution. A dashboard will summarize reports and provide basic statistics to identify recurring problems.

4. How the System Works

01	02	03	04	05
REPORT	**REGISTER**	**ASSIGN**	**UPDATE**	**RESOLVE**
Student submits issue	System creates issue ID	Admin assigns responsibility	Team updates status	Close issue and record result

Status flow: Reported → Assigned → In Progress → Resolved

5. Main Features

Issue reporting	Title, description, category, location and optional photo.
Issue tracking	Unique issue ID, current status and status history.
Admin dashboard	Review reports, assign responsibility, set priorities and update status.
Issue categories	Waste, water leakage, sanitation, electricity wastage and other campus issues.
Analytics	Counts of reported, pending, in-progress and resolved issues; recurring categories and resolution rate.

6. User Roles

STUDENT / STAFF	ADMIN / RESPONSIBLE TEAM
Submit an issue, add details and check its status.	Review reports, prioritize, assign and update progress.

7. Proposed Technology Stack

Component	Technology
Frontend	HTML, CSS and JavaScript
Authentication	Firebase Authentication
Database	Cloud Firestore
Photo storage (optional)	Firebase Storage
Charts (optional)	JavaScript charting library
Version control	Git and GitHub

8. Objectives

- Centralize reporting of campus environmental issues.
- Make issue status and responsibility easier to track.
- Help administrators prioritize and manage reports.
- Identify recurring problems through basic analytics.
- Evaluate usability and usefulness through a campus trial.

9. Testing and Evaluation

The prototype will be tested with a small group of campus users, subject to permission. The team will check:

- Whether reports can be submitted and stored correctly.
- Whether administrators can assign issues and update their status.
- Whether users can track progress and see accurate status information.
- Whether dashboard counts match the saved issue records.
- User feedback, and processing/resolution time where reliable timestamps are available.

10. Scope and Limitations

The first version will focus on reporting, administration, status tracking, and basic analytics. It will not automatically detect physical problems; users must report them and authorized personnel must update progress. Resolution depends on the participation of the responsible campus teams. Notifications, map visualization, and AI-assisted categorization may be considered for future versions.

11. Expected Outcomes

- A functional web application for reporting and tracking environmental issues.
- Organized assignments and resolution records.
- Better visibility into pending and recurring campus problems.
- Documented testing results, user feedback, limitations, and improvements.

12. Conclusion

The Environmental Issue Tracking System aims to provide a practical, affordable digital solution for managing campus environmental problems. By centralizing reporting, assignment, tracking, and analysis, it can make the process more organized and transparent. Its effectiveness will be assessed through campus testing and observed results.

Project status: Proposed — subject to team approval.